UNIVERSITÄT
LEIPZIG

Lecture

Topic Models

Introduction and Application

Leipzig, 17.12.2025

Dr. Ing. Andreas Niekler

Computational
Humanities
UNIVERSITÄT LEIPZIG

COVERED MATERIAL

OUTLINE

- Topic Model Intuition
  - Problems with frequentist approaches
  - The Model
- Applications
  - Results and Usage
  - Best practices

PROBLEMS WITH FREQUENTIST METHODS

FREQUENCY VS. CONTEXT

When we examine a word, or an n-gram, **out of the context** in which it appears, we inevitably **lose information** about how that word is being applied.

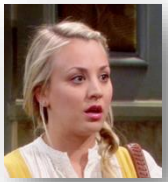
If we are merely interested in **stylistic habits**, such out-of-context counting can be usefully employed, but when it comes to drawing **semantic meaning** from a word, we require more than a count of that word‘s occurrence in the corpus. [1]

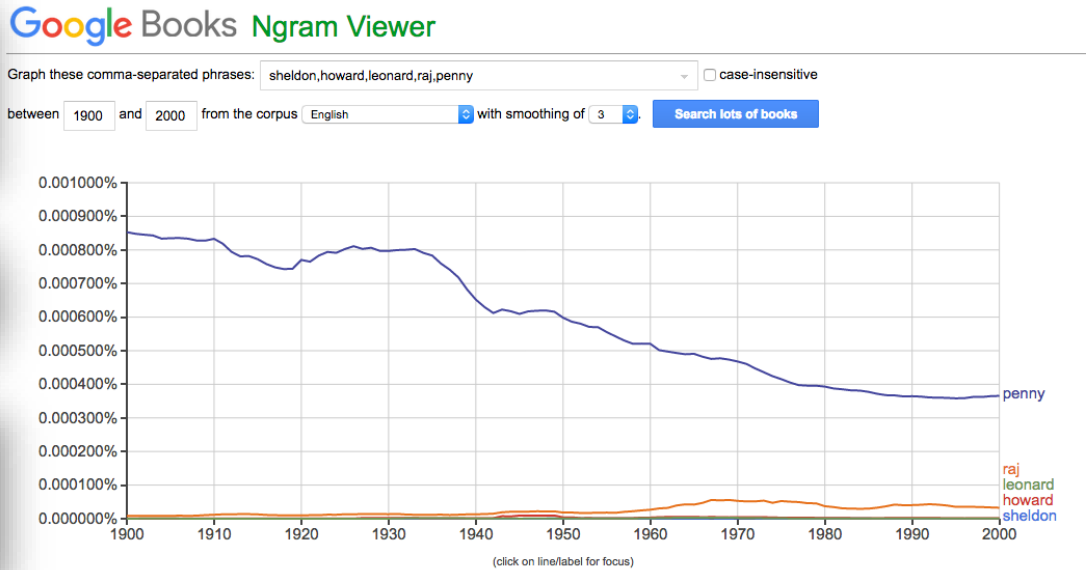
Remeber Firth: “You shall know a word by the company it keeps.“

[1] JOCKERS, M. L. (2013). *Macroanalysis: Digital Methods and Literary History*. University of Illinois Press. http://www.jstor.org/stable/10.5406/j.ctt2jcc3m

PROBLEMS WITH FREQUENTIST METHODS

FREQUENCY VS. CONTEXT





PROBLEMS WITH FREQUENTIST METHODS

FREQUENCY VS. CONTEXT

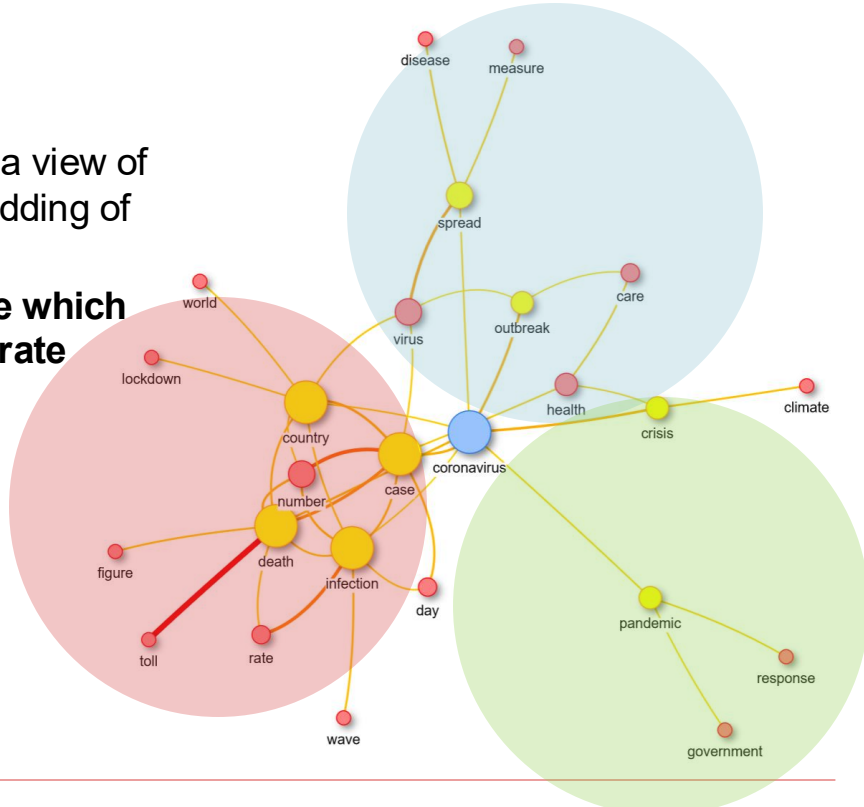
- Concordances show the context for a word, sentence by sentence … manual checking required ☹

	pre	*keyword*	*post*
1	be caused by a	**coronavirus**	, the family of
2	Chinese of a novel	**coronavirus**	emerging once more from
3	with new strain of	**coronavirus**	. Broadcasters look likely
4	new strain of the	**coronavirus**	, marking the first
5	to Sydney because of	**coronavirus**	outbreak " I saw
6	be about the unfolding	**coronavirus**	but he begins by
7	, similar to the	**coronavirus**	. Top medics ,
8	viruses . Although the	**coronavirus**	recently discovered in Wuhan
9	don't know where the	**coronavirus**	has come from -
10	Middle East plan and	**coronavirus**	. Climate emergency is
11	out of Wuhan in	**coronavirus**	evacuation A plane ,
12	the centre of the	**coronavirus**	outbreak , as officials
13	human-to-human transmission of the	**coronavirus**	in Europe , where
14	the spread of the	**coronavirus**	. Dean Smith lost
15	decision to frame the	**coronavirus**	on its front page
16	" inappropriately labelled the	**coronavirus**	by race " and
17	the source of the	**coronavirus**	being China . Mondays
18	. No backing for	**coronavirus**	claim The Daily Telegraph
19	fears that he had	**coronavirus**	" . The Daily
20	were afraid of the	**coronavirus**	is unsourced in the

PROBLEMS WITH FREQUENTIST METHODS

FREQUENCY VS. CONTEXT

- Cooccurrences give us a view of the semantic and embedding of single words ☹
- **We still need to decide which contexts may be separate senses**



PROBLEMS WITH FREQUENTIST METHODS

FREQUENCY VS. CONTEXT

If we may know the **sense of a word** by the **company of words** that surround it in a sentence, we may know a **theme** by the **sentences, paragraphs, chapters**, and even full books that express it.... What is needed in order to capture theme are **collocations of collocations** in a much larger scale. [1, p. 122]

[1] JOCKERS, M. L. (2013). *Macroanalysis: Digital Methods and Literary History*. University of Illinois Press. http://www.jstor.org/stable/10.5406/j.ctt2jcc3m

PROBLEMS WITH FREQUENTIST METHODS

TOPIC MODELS FOR THE RESCUE

A topic model takes a collection of texts as input. It discovers a set of “topics” — **recurring themes that are discussed in the collection** — and the **degree to which each document exhibits those topics**.

With probabilistic modeling for the humanities, the scholar can **build a statistical lens that encodes her specific knowledge, theories, and assumptions about texts**. She can then use that lens to examine and explore large archives of real sources. [2]

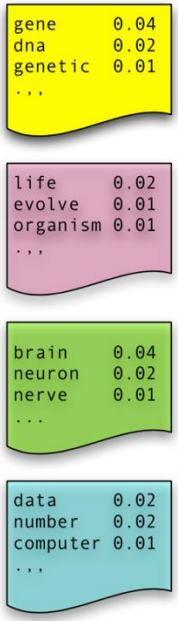
[2] David Blei *Topic Modeling and Digital Humanities Journal of Digital Humanities*. 2012
http://journalofdigitalhumanities.org/2-1/topic-modeling-and-digital-humanities-by-david-m-blei/

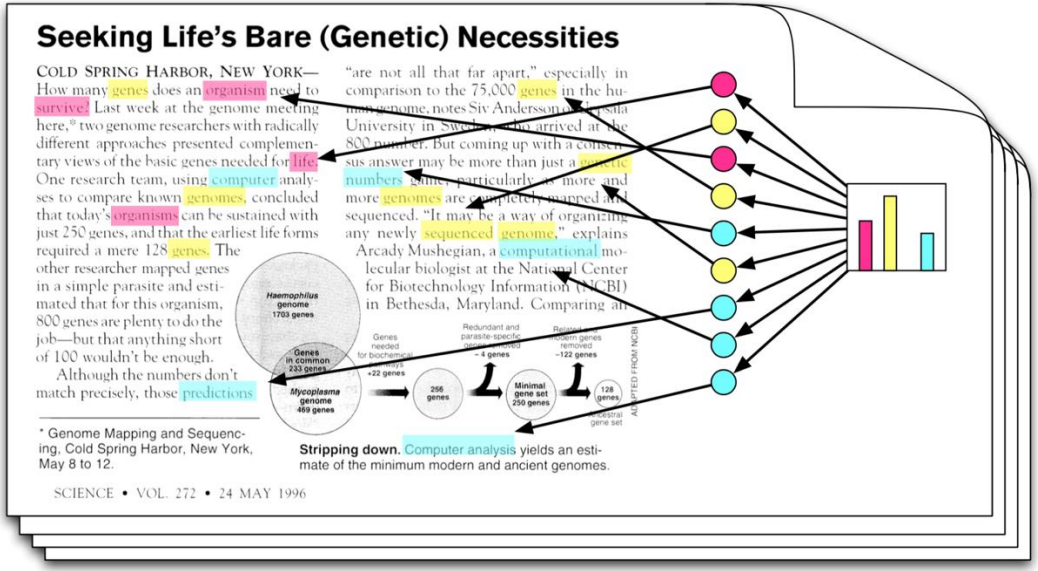
TOPIC MODELS

DETAILED MODEL AND INTUITION

- Topic modeling extracts topics from a text corpus based on patterns word usage.
- A topic is modeled as a list of words that co-occur in a statistically meaningful way.
  - Class of probabilistic graphical models which infer semantic coherences from large text collections as latent variables.
  - Latent variables = Topics
- Background Assumption:
  - Bag-of-words model as input data
  - Generative process (1. document author, 2. draw topic mixture, 3. draw terms from topics → document)
- 2 posterior distributions (Latent Variables)
  - θ : Documents = mixtures of topics
  - φ : Topics = probability distributions over terms

TOPIC MODELS





Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many genes does an organism need to survive? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those predictions "are not all that far apart," especially in comparison to the 75,000 genes in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a genetic numbers game, particularly as more and more genomes are completely mapped and sequenced. "It may be a way of organizing any newly sequenced genome," explains Arcady Mushegian, a computational molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an

* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.

SCIENCE • VOL. 272 • 24 MAY 1996

http://www.cs.columbia.edu/~blei/papers/Blei2012.pdf

- **Assumptions**
  - Each topic is a distribution over words.
  - Each text is a mixture of corpus-wide topics.
  - Each word is drawn from one of those topics.

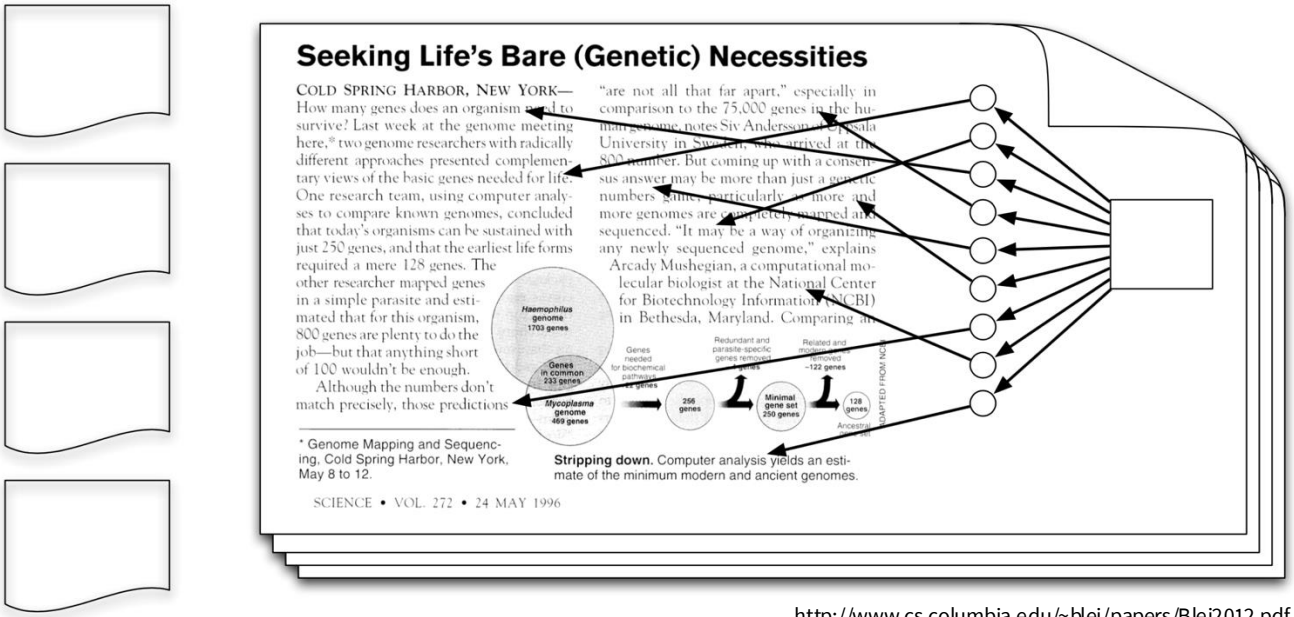
TOPIC MODELS

WHAT ARE „TOPICS“ IN TOPIC MODELING?

- ... Topic Modeling creates topical categories without a priori subject definitions.
- This may be the hardest concept to understand about topic modeling: unlike traditional classification systems where texts are fit into preexisting schemes, ... topic modeling determines the comprehensive list of subjects through its analysis of the word occurrences throughout a corpus of texts.
- The content of the documents – not a human indexer – determines the topics collectively found in those documents. [1]

[1] JOCKERS, M. L. (2013). *Macroanalysis: Digital Methods and Literary History*. University of Illinois Press. http://www.jstor.org/stable/10.5406/j.ctt2jcc3m

TOPIC MODELS



Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK— How many genes does an organism need to survive? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those predictions "are not all that far apart," especially in comparison to the 75,000 genes in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a genetic numbers game, particularly as more and more genomes are completely mapped and sequenced. "It may be a way of organizing any newly sequenced genome," explains Arcady Mushegian, a computational molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an

* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.

SCIENCE • VOL. 272 • 24 MAY 1996

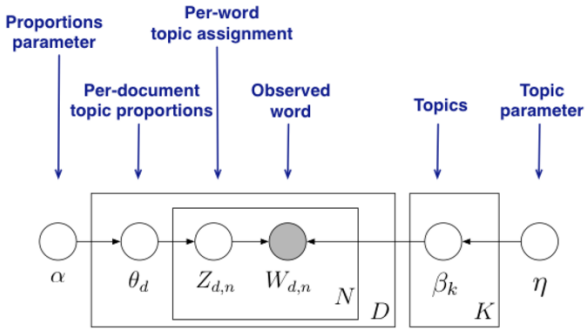
http://www.cs.columbia.edu/~blei/papers/Blei2012.pdf

- **Setting**
  - In reality, we observe only texts and words, not topics.
  - The aim of LDA/Topic Models is to infer the latent (say, hidden) topic structure.

TOPIC MODELS

LATENT DIRICHLET ALLOCATION

- The most popular technique is the **Latent Dirichlet Allocation (LDA)**
  - The terms topic modeling and LDA are often used synonymously.
  - The generative process can be seen as a graphical model



TOPIC MODELS

MAIN METHODOLOGICAL QUESTIONS

- **How many topics** should be found?
- **How many words** per topic?
- How should the text be **preprocessed**? (lemmatization, stop words out, POS tagging → only nouns?)
- **And**: What to do with topics that don't make sense – criteria?
  - topic coherence
  - topic interpretability

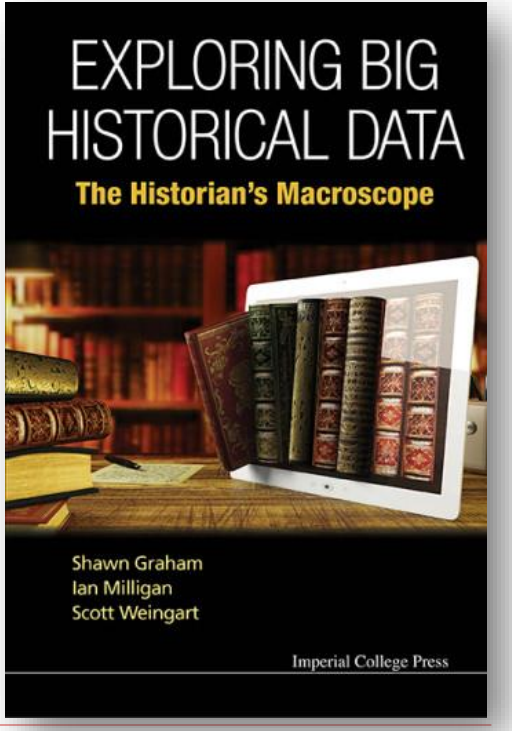
TOPIC MODELS

FURTHER READING

David Blei *Topic Modeling and Digital Humanities Journal of Digital Humanities*. 2012
http://journalofdigitalhumanities.org/2-1/topic-modeling-and-digital-humanities-by-david-m-blei/

TOPIC MODELS

FURTHER READING

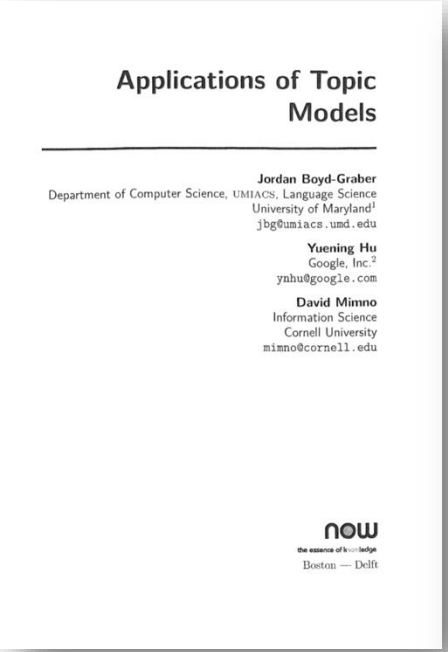


Graham, S., Milligan, I., Weingart, S., & Martin, K. (2022). *Exploring big historical data: The historian's macroscope* (Second edition). World Scientific.

TOPIC MODELS

FURTHER READING

Boyd-Graber, J., Hu, Y., & Mimno, D. (2017). Applications of Topic Models. *Foundations and Trends® in Information Retrieval*, *11*(2–3), 143–296. https://doi.org/10.1561/1500000030



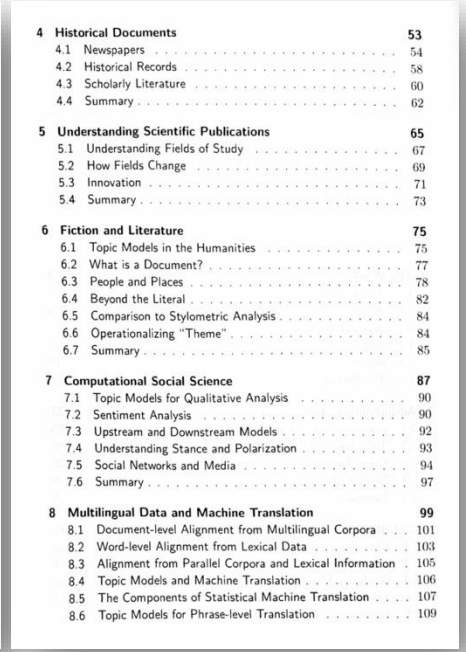
Applications of Topic Models

Jordan Boyd-Graber
Department of Computer Science, UMIACS, Language Science
University of Maryland[1]
jbg@umiacs.umd.edu

Yuening Hu
Google, Inc.[2]
ynhu@google.com

David Mimno
Information Science
Cornell University
mimno@cornell.edu

now
the essence of knowledge
Boston — Delft



4 Historical Documents 53
4.1 Newspapers 54
4.2 Historical Records 58
4.3 Scholarly Literature 60
4.4 Summary 62

5 Understanding Scientific Publications 65
5.1 Understanding Fields of Study 67
5.2 How Fields Change 69
5.3 Innovation 71
5.4 Summary 73

6 Fiction and Literature 75
6.1 Topic Models in the Humanities 75
6.2 What is a Document? 77
6.3 People and Places 78
6.4 Beyond the Literal 82
6.5 Comparison to Stylometric Analysis 84
6.6 Operationalizing "Theme" 84
6.7 Summary 85

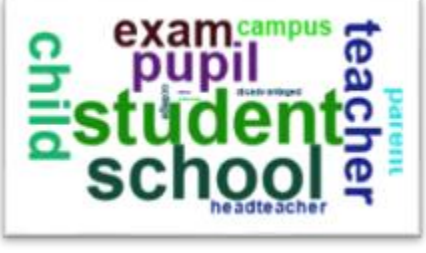
7 Computational Social Science 87
7.1 Topic Models for Qualitative Analysis 90
7.2 Sentiment Analysis 90
7.3 Upstream and Downstream Models 92
7.4 Understanding Stance and Polarization 93
7.5 Social Networks and Media 94
7.6 Summary 97

8 Multilingual Data and Machine Translation 99
8.1 Document-level Alignment from Multilingual Corpora 101
8.2 Word-level Alignment from Lexical Data 103
8.3 Alignment from Parallel Corpora and Lexical Information 105
8.4 Topic Models and Machine Translation 106
8.5 The Components of Statistical Machine Translation 107
8.6 Topic Models for Phrase-level Translation 109

TOPIC MODELS

EXAMPLE

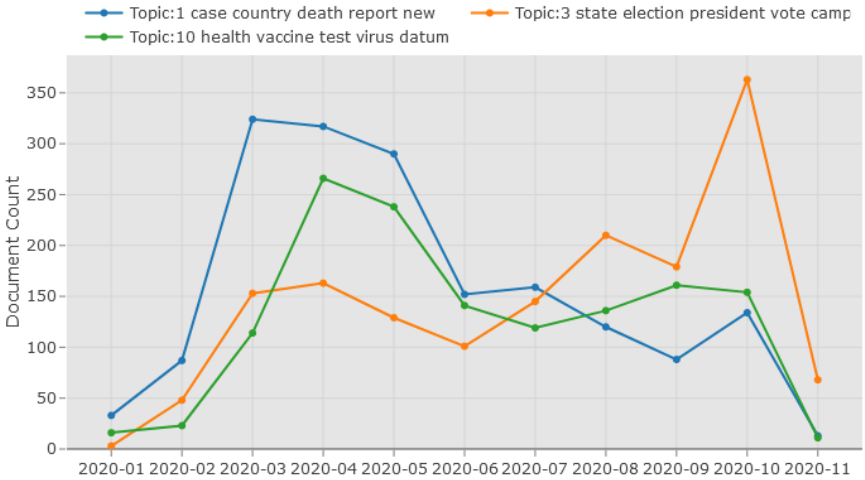
- **Corpus: 56400 Documents from the Guardian Newspaper containing the term „Coronavirus“ → K = 20**



TOPIC MODELS

EXAMPLE

- **Time Series based on Rank-1 document counts**

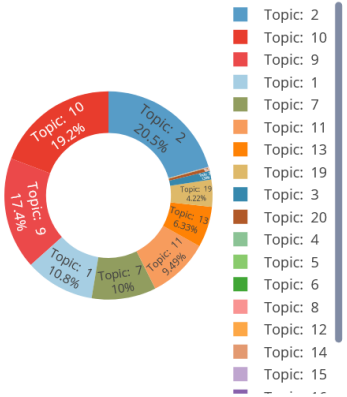


TOPIC MODELS

USE CASES IN DIGITAL HUMANITIES

– Validation

Distribution of topics for chosen document



Most relevant words for chosen topic

The Guardian view on Trump 's impeachment trial : the perils of outrageous immunity | Editorial . Chefs and busy home cooks swear by their store cupboard : it 's the lifeboat that carries you through when there 's only 10 minutes to throw dinner together . But are you making the best choices about what you keep in it ? . It 's no coincidence that internet phenomena share a vocabulary with certain diseases . As fast as a viral pathogen can spread in a world connected by air travel , bad information can move even faster . Very few people outside of China have been infected with the new coronavirus that was first detected late last year and was declared a global health emergency this week . But millions have already been exposed to false information about the virus . And while we may not be able to measure the damage that misinformation poses to society in the same way that we can take a patient 's temperature , it 's undeniable that the damage is real . Related : Wuhan , my hometown , is not a city of fear | Xiaoyu Lu Recent outbreaks of measles in countries that had previously eradicated the disease , but then become centers of anti - vaxx conspiracy mongering , are one concrete example of how disinformation can lead to concrete harm . Already , east Asians in Europe and the Americas are reporting incidents of racist hostility and exclusion . Such experiences will not be counted among the casualties of the disease , but will nevertheless leave a mark on their lives . Disinformation experts and fact - checkers are now working overtime to attempt to clean up the mess of honest misunderstandings , hoaxes and malicious lies that have spread across the internet . Naturally , attention has again turned to social media users and the social media companies whose platforms have played an important role in accelerating the frictionless spread of falsehood . It 's certainly fair to expect Facebook , Twitter , YouTube and TikTok , among others , to do everything they can to mitigate misinformation amid a global health emergency . Facebook and YouTube were incredibly negligent even as they grew into dominant providers of news and information , only agreeing to take proactive steps to address conspiracy theories about breaking news and health disinformation after facing considerable pressure from the press and politicians . But it 's also

TOPIC MODELS

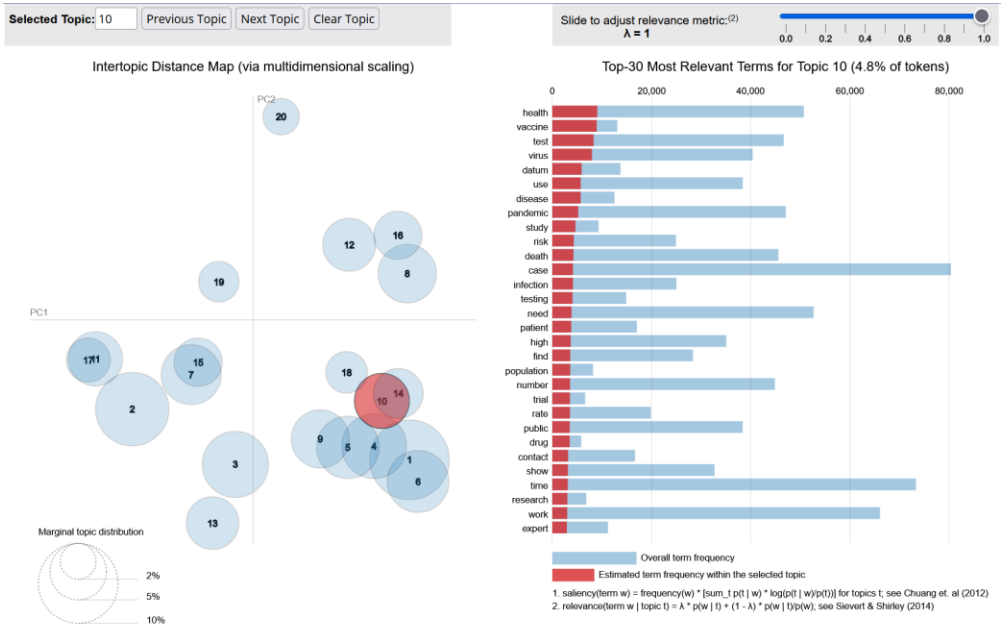
MANUAL EVALUATION

- **3 steps proposed by [10, 4, 9]:**
  1. Check semantic coherence of top N terms of each topic: Can you assign a topic label?
  2. Employ additional numeric measures of topic coherence to identify broad / incoherent topics
  3. Check if topic distribution over time complies with researcher intuition

[10] Evans, M. S. (2014). A Computational Approach to Qualitative Analysis in Large Textual Datasets. *PLoS ONE*, *9*(2), e87908. https://doi.org/10.1371/journal.pone.0087908
[4] Sievert, C., & Shirley, K. E. (2014). LDAvis: A method for visualizing and interpreting topics. *Proceedings of the workshop on interactive language learning, visualization, and interfaces*, 63–70. http://www.aclweb.org/website/old_anthology/W/W14/W14-31.pdf#page=73
[9] Chang, J., Boyd-Graber, J., Gerrish, S., Wang, C., & Blei, D. (2009). Reading tea leaves: How humans interpret topic models. *Neural Information Processing Systems*, 1–9.

TOPIC MODELS

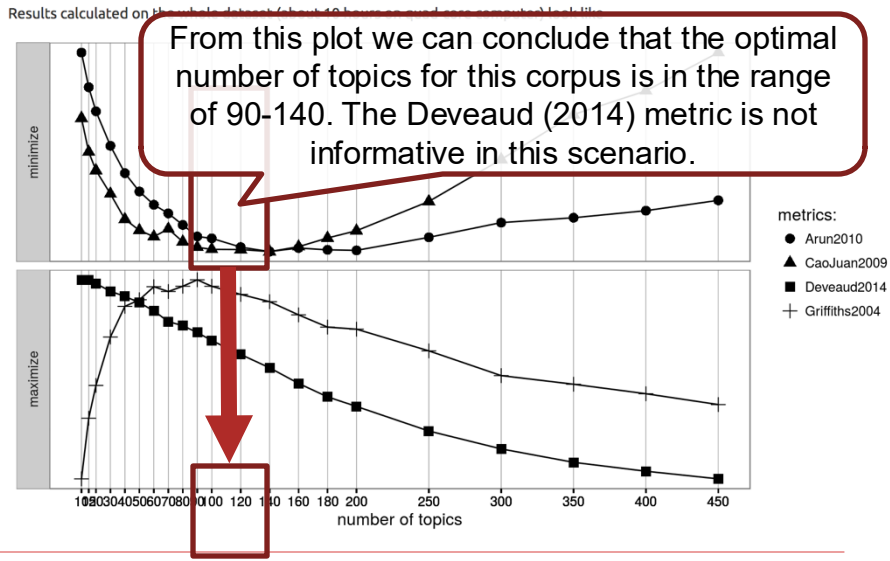
LDAVIS



TOPIC MODELS

MODEL SELECTION AND EVALUATION

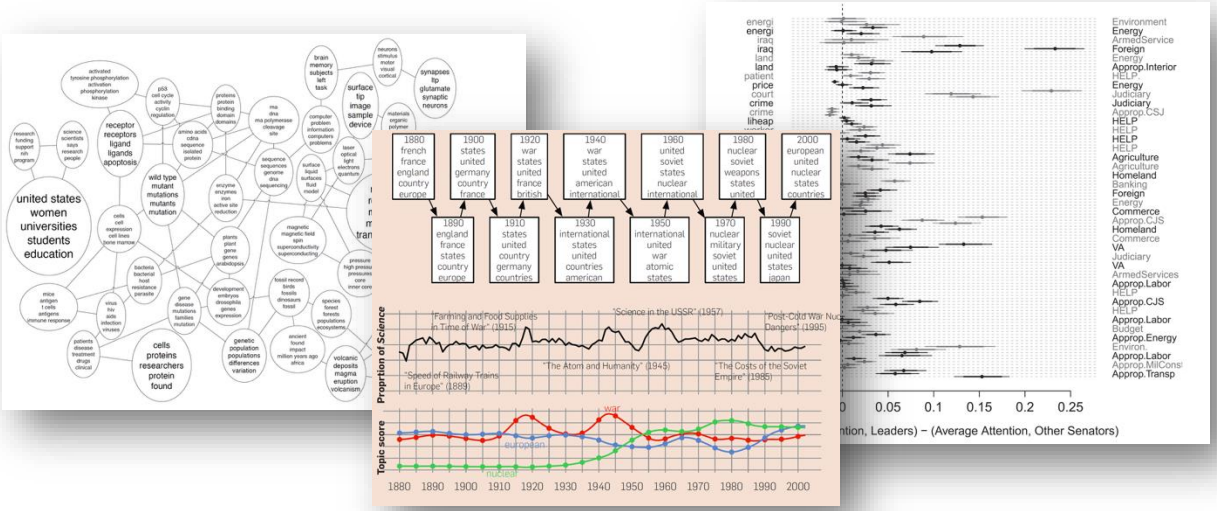
- LDATuning in R: Select optimal number of topics for the LDA Model based on informed evaluation measures (http://rpubs.com/siri/ldatuning)
- Calculate different models and aquire metrics
- Look for the optimal K by observing the optimal values
- See Literature for the papers explaining the measures



TOPIC MODELS

ALTERATIONS AND VARIANTS

- Extentions are Author Topic Model (ATM) [Rosen-Zvi et al. 2004], Correlated Topic Model (CTM) [Blei & Lafferty 2007], Relational Topic Model (RTM) [Chang, Blei 2010] and Structural Topic Model (STM) [Roberts et. al. 2014]



APPROACHES

LITERATURE

Rajkumar Arun, V. Suresh, C. E. Veni Madhavan, and M. N. Narasimha Murthy. 2010. On finding the natural number of topics with latent dirichlet allocation: Some observations. In *Advances in knowledge discovery and data mining*, Mohammed J. Zaki, Jeffrey Xu Yu, Balaraman Ravindran and Vikram Pudi (eds.). Springer Berlin Heidelberg, 391–402. http://doi.org/10.1007/978-3-642-13657-3_43

Cao Juan, Xia Tian, Li Jintao, Zhang Yongdong, and Tang Sheng. 2009. A density-based method for adaptive lDA model selection. *Neurocomputing — 16th European Symposium on Artificial Neural Networks 2008* 72, 7–9: 1775–1781. http://doi.org/10.1016/j.neucom.2008.06.011

Romain Deveaud, Éric SanJuan, and Patrice Bellot. 2014. Accurate and effective latent concept modeling for ad hoc information retrieval. *Document numérique* 17, 1: 61–84. http://doi.org/10.3166/dn.17.1.61-84

Thomas L. Griffiths and Mark Steyvers. 2004. Finding scientific topics. *Proceedings of the National Academy of Sciences* 101, suppl 1: 5228–5235. http://doi.org/10.1073/pnas.0307752101

Blei, D. (2012). Topic modeling and the digital humanities. *Journal of Digital Humanities*, 2(1), 2-1-.

Blei, D. M. (2012). Probabilistic topic models. *Communications of the ACM*, *55*(4), 77–84. https://doi.org/10.1145/2133806.2133826

Boyd-Graber, J., Hu, Y., & Mimno, D. (2017). Applications of Topic Models. *Foundations and Trends® in Information Retrieval*, *11*(2–3), 143–296. https://doi.org/10.1561/1500000030

Chang, J., Boyd-Graber, J., Gerrish, S., Wang, C., & Blei, D. (2009). Reading tea leaves: How humans interpret topic models. *Neural Information Processing Systems*, 1–9.

Evans, M. S. (2014). A Computational Approach to Qualitative Analysis in Large Textual Datasets. *PLoS ONE*, *9*(2), e87908. https://doi.org/10.1371/journal.pone.0087908

JOCKERS, M. L. (2013). *Macroanalysis: Digital Methods and Literary History*. University of Illinois Press. http://www.jstor.org/stable/10.5406/j.ctt2jcc3m

Kemman, M., Kleppe, M., & Scagliola, S. (2013). Just Google It Digital Research Practices of Humanities Scholars. *Cornell Digital Libraries*, 1–17.

Koltcov, S., Nikolenko, S. I., Koltsova, O., & Bodrunova, S. (2016). Stable Topic Modeling for Web Science: Granulated LDA. *Proceedings of the 8th ACM Conference on Web Science*, 342–343. https://doi.org/10.1145/2908131.2908184

Lancichinetti, A., Sirer, M. I., Wang, J. X., Acuna, D., Körding, K., & Amaral, L. A. N. (2015). High-Reproducibility and High-Accuracy Method for Automated Topic Classification. *Phys. Rev. X*, *5*(1), 011007. https://doi.org/10.1103/PhysRevX.5.011007

Maier, D., Waldherr, A., Miltner, P., Wiedemann, G., Niekler, A., Keinert, A., Pfetsch, B., Heyer, G., Reber, U., Häussler, T., Schmid-Petri, H., & Adam, S. (2018). Applying LDA Topic Modeling in Communication Research: Toward a Valid and Reliable Methodology. *Communication Methods and Measures*, *12*(2–3), 93–118. https://doi.org/10.1080/19312458.2018.1430754

Mimno, D., Wallach, H. M., Talley, E., Leenders, M., & McCallum, A. (2011). Optimizing Semantic Coherence in Topic Models. *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 262–272.

Niekler, A., & Jähnichen, P. (2012). Matching Results of Latent Dirichlet Allocation for Text. *Proceedings of ICCM 2012, 11th International Conference on Cognitive Modeling*, 317–322.

Wallach, H. M., Murray, I., Salakhutdinov, R., & Mimno, D. (2009). Evaluation methods for topic models. *Proceedings of the 26th Annual International Conference on Machine Learning - ICML '09*, 1–8. https://doi.org/10.1145/1553374.1553515